Michalis Karaoulas

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GitHub: github.com/karaoulas | **Portfolio:** codelikemike.me **Linkedin:** https://www.linkedin.com/in/karaoulas/

EDUCATION

BSc Computer Science Aristotle University of Thessaloniki (AUTH)

MSc in Applied Data Science (Attending)

Relevant Courses: Software Engineering, Object-Oriented Programming, Distributed Systems, Cloud Technologies,Data structures.

TECHNICAL SKILLS

Data Science & AI: Pandas, NumPy, scikit-learn, SpaCy, Machine Learning

Web Technologies: Flask, HTML/CSS, PHP, Web Scraping (Selenium, BeautifulSoup), SPARQL

Databases & Data Engineering: SQL, MySQL, PostgreSQL, Data Extraction pipelines

Languages: Python,C,C++

Cloud & DevOps: AWS (EC2, S3), Docker, GitHub Actions, Linux

Certifications: AWS Certified AI Practitioner,

AWS Certified Cloud Practitioner (Expected)

LANGUAGES

- **Greek:** Native
- **English:** C2 (Proficient)
- **German:** B1(Intermediate)

CORE INTERESTS

- Data Science & Applied Machine Learning
- Data engineering
- Natural Language Processing (NLP)
- Data Pipelines
- Software Engineering

PROFESSIONAL EXPERIENCE

IT Specialist Hellenic Army – IT Research Office | Nov 2025 – Oct 2026

- Act as the primary IT and software systems support specialist within the military headquarters operations.
- Designed and deployed an automated data-entry platform utilizing QR codes and spreadsheet synchronization to optimize recruit onboarding workflows.
- Configured and maintained a secure Local Area Network (LAN) deployment, ensuring high-availability resource sharing across the infrastructure.

Software Engineering Intern Web2learn | May 2025 – Oct 2025

- Developed backend authentication modules integrating multiple web platforms with a central user directory.
- Implemented secure Single Sign-On (SSO) flows using Keycloak with OAuth 2.0 and OIDC protocols.
- Managed secure JWT token lifecycles and implemented strict endpoint access controls during application runtime.

Insurance Office Assistant Insurance Office | Nov 2022 – Aug 2023

- Managed database workflows and streamlined internal business data operations.
- Designed custom Python analytics scripts to parse, automate, and extract insights from complex structural contract data, significantly reducing manual data-entry times.

KEY PROJECTS

BSc Thesis: NLP Curricula Extraction & Skill Gap Analysis *(Feb 2026)*

- Built an automated text processing data pipeline in Python to ingest unstructured educational documents and benchmark them against market indicators.
- Utilized text mining utilities (BeautifulSoup, SpaCy) for tokenization, lemmatization, and customized stop-word filtering.
- Implemented Machine Learning classification using K-means clustering and Latent Dirichlet Allocation (LDA) Topic Modeling.

Distributed Messaging System

- Designed and implemented a multithreaded client-server communications application using Java RMI.
- Structured object-oriented server logic to manage concurrent socket threads, integrating token verification for secure sessions.

MySQL Database System

- Designed and implemented a relational database from scratch, including complete ER diagram modeling, table relationships, schema constraints, and complex SQL queries.

End-to-End Medallion Data Lakehouse

- Built an end-to-end Medallion Data Lakehouse architecture using Databricks, PySpark, and SQL for scalable data ingestion, transformation, and analytical modeling across Bronze, Silver, and Gold layers.